

Load Non-Decomposition: Invariance of Load in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

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April 15, 2026 — Version 1

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Abstract

This document defines the non-decomposition condition governing Load (L) in Dimensional Human Field Theory (DHFT).

Load is a single invariant variable. Subdivision of Load into types, categories, or domains does not constitute a valid system description.

I. Scope

This document establishes the invariance of Load within DHFT.

It constrains all valid system descriptions to a single Load variable and excludes decomposition into multiple forms within the minimal system layer.

II. Definition

Load (L) is defined as total demand acting on the system.

Load is cumulative, transferable, and time-dependent. It is not subdivided into types, categories, or domains within the minimal system.

III. Non-Decomposition Condition

Load is not decomposed.

Any representation of Load as multiple distinct types is a non-admissible system description.

IV. Constraint

Valid system descriptions preserve Load as a single invariant variable.

Subdivision of Load introduces additional variables and violates the minimal constraint model.

V. Exclusion

The following do not constitute admissible representations of Load at the minimal system layer:

- Type-based distinctions
- Domain-specific classifications
- Interpretive categories
- Context-dependent subdivisions

These introduce non-invariant structure and are excluded from the minimal system.

VI. Relation to System Structure

Load (L) operates as a single variable in relation to Capacity (C) and Boundary (B) as a single variable.

All system states and relations are defined over the invariant set $\{L, C, B\}$.

VII. Statement

Load is a single invariant variable. It is not decomposed.

All valid system descriptions preserve Load as non-decomposed within the minimal constraint model.

VIII. System Reference

The system defined in this document is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All valid system descriptions reduce to Load (L), Capacity (C), and Boundary (B).

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical series.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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